

Ghassen Bargougui

Application Engineer | Java Applications, Cloud & Platform Engineering

Fukuoka, Japan · +81 70 8502 9039 · ghassen.brg16@gmail.com
ghassen.io · linkedin.com/in/ghassenbrg · github.com/ghassenbrg

Profile

Application Engineer with seven years of Java and full-stack experience across banking, insurance, and card payments in Tunisia, Luxembourg, and Japan. At Rakuten (Technology Incubation Team, Rakuten Card systems), my work spans hybrid-cloud migration, application and infrastructure design, Kubernetes delivery, CI/CD, performance testing, and observability. Creator of the Slide Agent, Pockito, and SubMate personal projects.

Professional experience

Application Engineer - Rakuten · Fukuoka, Japan | Dec 2025 - Present

In the Technology Incubation Team, I work on Rakuten Card systems across Java application development and cloud/platform engineering, and help lead a hybrid-cloud migration alongside my manager and project manager, coordinating implementation with development, infrastructure, CI/CD, and security teams.

- Upgraded the e-Navi API to JDK 25 and Helidon 4.3, resolving dependency and CDI issues, updating tests, and documenting the handover.
- Designed and validated Kubernetes deployments for the JSF, API, and Engine modules as part of the hybrid-cloud migration, including services, environment configuration, and Filestore mounts.
- Run performance tests with k6 and JMeter, diagnose bottlenecks, and address them through application-design changes or resource adjustments.
- Reworked application logging as structured JSON for Cloud Logging (Log4j2 with a custom formatter); currently helping move file-based application and trace logs to Kafka/Confluent.
- Develop Java APIs with Redis-backed session management, service-to-service calls, and LLM API integration on Kubernetes infrastructure.
- Resolved deployment, SSL, and GCP-to-on-premises connectivity issues with infrastructure and security teams, and traced a Kafka messaging issue to an incorrect CDI bean scope.
- Built a GitHub Actions pipeline for builds, Docker image publishing to Artifact Registry, and GKE deployments.

Software Engineer - Sogeti · Client assignment at BGL BNP Paribas · Luxembourg | Feb 2025 - Nov 2025

Developed and maintained internal banking applications for BGL BNP Paribas, including customer mandate management for retail banking.

- Delivered customer mandate management features with Spring Boot, Spring Batch, and Angular Material.
- Developed REST APIs and Hibernate data access for internal applications on PostgreSQL and Oracle.
- Automated builds and deployments with Maven and Jenkins, and maintained code quality through unit tests, SonarQube analysis, and peer reviews.

Java Full-Stack Engineer - CBTW · Luxembourg | Sep 2024 - Jan 2025

Modernized Java services and developed web applications with Spring Boot, Angular, and PostgreSQL.

- Refactored monolithic application components into microservices as part of the Java service modernization.
- Built and documented REST APIs for service integration, covering versioning and security requirements.
- Automated deployments with Jenkins and Docker, reducing manual release steps.

Analyst Developer - VERMEG for Banking and Insurance · Tunis, Tunisia | Jan 2023 - Aug 2024

Software Developer - VERMEG for Banking and Insurance · Tunis, Tunisia | Oct 2019 - Dec 2022

Built and maintained backend services for insurance and financial applications, then took on functional analysis and cross-team solution design as Analyst Developer.

- Improved API response times through query optimization and caching.
- Implemented Keycloak authentication and role-based access control, and managed CI/CD pipelines and rollback strategies.
- Responsible for assigned backend components; worked with business, architecture, and framework teams on requirements, microservices, and API contracts.
- Documented designs in UML and sequence diagrams, and acted as a primary contact for cross-team design questions.

Selected company projects

- **SalesFlow** (2023 - 2024): Led the migration of insurance workflows to a Camunda BPM-based architecture to reduce manual processing. Integrated ActiveMQ to automate bulk-processing workflows.
- **PackManager** (2022 - 2023): Profiled Java execution and memory usage with JProfiler and optimized the identified bottlenecks.

Other company projects: Solife Digital & Generali Portal, Magikforms, Insurance Pricing Module (2020 - 2021)

Technical skills

Java Application Engineering: Java, Spring Boot, Helidon, MicroProfile, Hibernate, RESTful API Design, OpenAPI, Maven

Cloud Delivery & CI/CD: Kubernetes, Google Cloud Platform (GCP), Google Kubernetes Engine (GKE), Docker, GitHub Actions, Jenkins

Performance & Observability: k6, JMeter, JProfiler, Cloud Logging, Log4j2

Data & Messaging: PostgreSQL, Oracle (PL/SQL), Redis, Kafka / Confluent, ActiveMQ

Frontend, Mobile & Browser: Angular, TypeScript, Vue.js, Nuxt, Node.js, Flutter, Dart, Chrome Extensions (MV3)

AI & Agent Tooling: Model Context Protocol (MCP), LLM API Integration

Quality & Security: JUnit, Mockito, SonarQube, Keycloak (OAuth2, OIDC, JWT), Git

Personal projects

Slide Agent · Creator & Maintainer · Open source · npm package | 2026

Open-source engine and MCP tooling for generating editable PowerPoint decks with native charts and layout validation.

- Built a TypeScript engine that compiles a grid-based composition language into native PowerPoint with editable charts and tables, theme colors, and embedded fonts.
- Published the engine on npm as `@slide-agent/core`, with a CLI, library API, and MCP server; template filling generates decks from CSV data without model calls.

Technologies: TypeScript, Node.js, MCP, OOXML, LLM APIs, Vitest, GitHub Actions · slide-agent.ghassen.io · github.com/ghassenbrg/slide-agent

Pockito · Creator & Full-Stack Engineer · Personal project | 2025 - 2026

Personal and shared finance app: Flutter and Nuxt clients, Spring Boot service with REST and MCP interfaces.

- Designed a Spring Boot backend in which REST and MCP adapters share one domain service, and built the Flutter and Nuxt clients.
- Built Kubernetes deployment and rollback scripts with Traefik routing, plus automated tests and end-to-end release checks.

Technologies: Java 25, Spring Boot, Flutter, Nuxt, PostgreSQL, Redis, Keycloak, Kubernetes, MCP · pockito.ghassen.io

SubMate · Creator & Maintainer · Open source · Chrome extension | 2026

Chrome extension for synchronized subtitle translation on Netflix, TVer, and Prime Video, translating on-device by default with optional cloud providers.

- Built a Manifest V3 extension on Chrome's on-device Translator API, with local caching and no SubMate backend, account, or telemetry.
- Added optional cloud translation using the user's own API key, and localized the interface in English, Japanese, French, and Arabic with right-to-left support.

Technologies: TypeScript, Chrome Extensions (MV3), Translator API, LLM APIs, IndexedDB, Vitest · submate.ghassen.io · github.com/ghassenbrg/SubMate

Other personal projects

- **Orbit Ways:** Browser-based multiplayer strategy board game with real-time game state over WebSockets. ghassen.io/orbitways
- **Traffic Forward** (2026): Spring Boot HTTP forwarding tool with configurable routing and request/response diagnostics. github.com/ghassenbrg/traffic-forward

Education, certification & languages

Engineering Degree (Diplôme d'ingénieur) in Software Architecture · ESPRIT (Private Higher School of Engineering and Technology), Tunis, Tunisia · Sep 2019 - Jul 2023

Bachelor's Degree in Computer Science · ISIMM (Higher Institute of Computer Science and Mathematics), Monastir, Tunisia · Sep 2015 - Jun 2019

Oracle Certified Professional, Java SE 11 Developer · Oracle · Sep 2022 · Oracle CertView: bit.ly/ocp11gb

Languages: Arabic (Native), English (Fluent), French (Fluent), Japanese (Basic)